ByteChat - 100 Functional Test Cases

Application: ByteChat - Real-Time Chat Application
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Test Case Template

Each test case follows this structure:
- **Test Case ID**: Unique identifier
- **Test Case Title**: Brief description
- **Priority**: High/Medium/Low
- **Preconditions**: Requirements before test
- **Test Steps**: Step-by-step instructions
- **Expected Result**: What should happen
- **Actual Result**: To be filled during testing
- **Status**: Pass/Fail/Blocked

User Authentication & Registration

TC-001: User Registration with Valid OTP
Priority: High
Preconditions: User has valid email address
Test Steps:
1. Navigate to registration page
2. Enter email: test@example.com
3. Click "Send OTP"
4. Check email for OTP code
5. Enter received OTP code
6. Enter password: Test@123
7. Enter display name: John Doe
8. Click "Register"

Expected Result:
- User account created successfully
- JWT tokens received
- User redirected to landing page
- Success message displayed

TC-002: User Registration with Invalid OTP
Priority: High

Preconditions: User on registration page
Test Steps:
1. Enter email: test@example.com
2. Click "Send OTP"
3. Enter incorrect OTP: 000000
4. Enter password and display name
5. Click "Register"

Expected Result:
- Registration fails
- Error message: "Invalid or expired OTP"
- User remains on registration page

TC-003: User Registration with Expired OTP
Priority: Medium
Preconditions: User has requested OTP
Test Steps:
1. Request OTP for email
2. Wait for OTP to expire (>10 minutes)
3. Enter expired OTP
4. Complete registration form
5. Click "Register"

Expected Result:
- Registration fails
- Error message: "OTP has expired"
- Option to resend OTP displayed

TC-004: User Login with Valid Credentials
Priority: High
Preconditions: User account exists
Test Steps:
1. Navigate to login page
2. Enter email: test@example.com
3. Enter password: Test@123
4. Click "Login"

Expected Result:
- Login successful
- JWT tokens received and stored
- User redirected to landing page
- User's workspaces displayed

TC-005: User Login with Invalid Password
Priority: High
Preconditions: User account exists
Test Steps:
1. Navigate to login page
2. Enter valid email
3. Enter incorrect password
4. Click "Login"

Expected Result:
- Login fails
- Error message: "Invalid credentials"
- User remains on login page

TC-006: User Login with Non-existent Email
Priority: High
Preconditions: None
Test Steps:
1. Navigate to login page
2. Enter non-existent email: notfound@example.com
3. Enter any password
4. Click "Login"

Expected Result:
- Login fails
- Error message: "User not found" or "Invalid credentials"
- User remains on login page

TC-007: JWT Token Refresh
Priority: High
Preconditions: User logged in, access token about to expire
Test Steps:
1. User logged in with valid session
2. Wait for access token to expire (15 minutes)
3. Make any API request
4. Observe token refresh mechanism

Expected Result:
- Access token automatically refreshed
- New access token received
- Original request completed successfully
- No user interruption

TC-008: Logout Functionality
Priority: High
Preconditions: User logged in
Test Steps:
1. User logged in and on any page
2. Click user profile icon
3. Click "Logout" button
4. Confirm logout

Expected Result:
- User logged out successfully
- Tokens cleared from storage
- Redirected to login page
- Cannot access protected routes

TC-009: Password Reset - Request OTP
Priority: Medium
Preconditions: User account exists
Test Steps:
1. Navigate to login page
2. Click "Forgot Password"
3. Enter registered email
4. Click "Send OTP"
5. Check email

Expected Result:
- OTP sent to email

- Success message displayed
- OTP input field appears

TC-010: Password Reset - Complete Flow
Priority: Medium
Preconditions: User has received password reset OTP
Test Steps:
1. Enter received OTP
2. Enter new password: NewPass@123
3. Confirm new password: NewPass@123
4. Click "Reset Password"
5. Try logging in with new password

Expected Result:
- Password reset successful
- Success message displayed
- Can login with new password
- Old password no longer works

TC-011: Registration with Duplicate Email
Priority: High
Preconditions: User account already exists with email
Test Steps:
1. Navigate to registration page
2. Enter existing email
3. Request OTP
4. Complete registration form

Expected Result:
- Registration fails
- Error message: "Email already registered"
- Suggestion to login instead

TC-012: Registration with Weak Password
Priority: Medium
Preconditions: None
Test Steps:
1. Navigate to registration page
2. Enter valid email and OTP
3. Enter weak password: 123
4. Try to register

Expected Result:
- Validation error displayed
- Message: "Password must be at least 8 characters"
- Registration blocked until valid password

TC-013: Session Persistence After Browser Refresh
Priority: Medium
Preconditions: User logged in
Test Steps:
1. User logged in successfully
2. Refresh browser page (F5)
3. Observe user state

Expected Result:

- User remains logged in
- Session persisted
- User data still available
- No redirect to login

TC-014: Multiple Login Sessions
Priority: Low
Preconditions: User account exists
Test Steps:
1. Login from Browser 1
2. Login from Browser 2 with same credentials
3. Perform actions in both browsers

Expected Result:
- Both sessions active
- Actions reflected in both browsers
- No session conflict

TC-015: Login with Email Case Insensitivity
Priority: Low
Preconditions: User registered with test@example.com
Test Steps:
1. Navigate to login page
2. Enter email: TEST@EXAMPLE.COM
3. Enter correct password
4. Click "Login"

Expected Result:
- Login successful
- Email treated as case-insensitive
- User logged in normally

Workspace Management

TC-016: Create Workspace with Valid OTP
Priority: High
Preconditions: User logged in
Test Steps:
1. Click "Create Workspace" button
2. Enter email for verification
3. Click "Send OTP"
4. Enter received OTP
5. Enter workspace name: "My Team"
6. Enter description: "Team workspace"
7. Select privacy: Public
8. Click "Create"

Expected Result:
- Workspace created successfully
- User set as workspace owner
- Default #general channel created
- User added to #general channel
- Redirected to workspace chat

TC-017: Create Private Workspace
Priority: High

Preconditions: User logged in, OTP verified
Test Steps:
1. Navigate to workspace creation
2. Complete OTP verification
3. Enter workspace name
4. Select "Private" option
5. Click "Create"

Expected Result:
- Private workspace created
- Only visible to members
- Owner can invite users
- Privacy setting saved correctly

TC-018: View All User Workspaces
Priority: High
Preconditions: User is member of multiple workspaces
Test Steps:
1. Login to application
2. Navigate to landing page
3. Observe workspace list

Expected Result:
- All workspaces displayed
- Shows owned and member workspaces
- Workspace names and descriptions visible
- Click to enter workspace

TC-019: Invite User to Workspace by Email
Priority: High
Preconditions: User is workspace owner/admin
Test Steps:
1. Open workspace
2. Click "Invite Members" in sidebar
3. Enter invitee email: newuser@example.com
4. Click "Send Invitation"

Expected Result:
- Invitation created with PENDING status
- Notification sent to invitee
- Success message displayed
- Invitation appears in invitee's notifications

TC-020: Accept Workspace Invitation
Priority: High
Preconditions: User has pending workspace invitation
Test Steps:
1. Login as invited user
2. Click notification bell icon
3. View workspace invitation
4. Click "Accept" button

Expected Result:
- Invitation status changed to ACCEPTED
- User added to workspace as MEMBER
- User added to #general channel
- Workspace appears in user's workspace list
- Notification marked as read

TC-021: Reject Workspace Invitation
Priority: Medium
Preconditions: User has pending workspace invitation
Test Steps:
1. Login as invited user
2. Open notifications panel
3. View workspace invitation
4. Click "Reject" button

Expected Result:
- Invitation status changed to REJECTED
- User not added to workspace
- Notification removed from list
- Can still be invited again later

TC-022: Leave Workspace as Member
Priority: Medium
Preconditions: User is workspace member (not owner)
Test Steps:
1. Open workspace
2. Click workspace settings
3. Click "Leave Workspace"
4. Confirm action

Expected Result:
- User removed from workspace
- User removed from all workspace channels
- Workspace no longer visible to user
- Confirmation message displayed

TC-023: Archive Workspace
Priority: Medium
Preconditions: User is workspace owner
Test Steps:
1. Open workspace settings
2. Click "Archive Workspace"
3. Confirm action

Expected Result:
- Workspace archived (isArchived = true)
- Workspace moved to archived section
- No new channels can be created
- Existing content remains accessible
- Can be restored later

TC-024: View Workspace Members
Priority: Medium
Preconditions: User is workspace member
Test Steps:
1. Open workspace
2. Click "Members" or workspace name
3. View member list

Expected Result:
- All workspace members displayed

- Shows display name and avatar
- Shows member roles (OWNER/ADMIN/MEMBER)
- Shows online status

TC-025: Remove Member from Workspace
Priority: High
Preconditions: User is workspace owner
Test Steps:
1. Open workspace members list
2. Select a member
3. Click "Remove Member"
4. Confirm action

Expected Result:
- Member removed from workspace
- Member removed from all channels
- Member loses access immediately
- Workspace no longer visible to removed user

TC-026: Transfer Workspace Ownership
Priority: Low
Preconditions: User is workspace owner, another admin exists
Test Steps:
1. Open workspace settings
2. Click "Transfer Ownership"
3. Select new owner from admin list
4. Confirm transfer

Expected Result:
- Ownership transferred to selected user
- Previous owner becomes admin
- New owner has full control
- Change reflected immediately

TC-027: Create Workspace with Duplicate Name
Priority: Low
Preconditions: Workspace with name already exists
Test Steps:
1. Try to create workspace
2. Enter existing workspace name
3. Complete form
4. Click "Create"

Expected Result:
- Workspace created successfully
- Duplicate names allowed
- Each workspace has unique ID
- No conflict

TC-028: Workspace with Special Characters in Name
Priority: Low
Preconditions: User logged in
Test Steps:
1. Create new workspace
2. Enter name: "Team @#$% 2024"
3. Complete creation

Expected Result:
- Workspace created successfully
- Special characters handled properly
- Name displayed correctly
- No encoding issues

TC-029: View Workspace Without Membership
Priority: Medium
Preconditions: Private workspace exists, user not a member
Test Steps:
1. Try to access workspace URL directly
2. Attempt to view workspace

Expected Result:
- Access denied
- Error message: "You don't have access"
- Redirected to landing page
- Workspace not visible in list

TC-030: Workspace Pagination
Priority: Low
Preconditions: User is member of 50+ workspaces
Test Steps:
1. Navigate to landing page
2. Scroll through workspace list
3. Observe pagination

Expected Result:
- Workspaces loaded in pages (50 per page)
- Smooth scrolling
- Load more on scroll
- No performance issues

Channel Management

TC-031: Create Public Channel
Priority: High
Preconditions: User is workspace member
Test Steps:
1. Open workspace
2. Click "Create Channel" button
3. Enter channel name: "announcements"
4. Enter description: "Team announcements"
5. Select "Public"
6. Click "Create"

Expected Result:
- Channel created successfully
- Creator added as channel owner
- Channel visible to all workspace members
- Channel appears in sidebar

TC-032: Create Private Channel
Priority: High
Preconditions: User is workspace member

Test Steps:
1. Click "Create Channel"
2. Enter channel name: "private-team"
3. Select "Private"
4. Click "Create"

Expected Result:
- Private channel created
- Only creator can see initially
- Lock icon displayed
- Can invite specific members

TC-033: Invite User to Private Channel
Priority: High
Preconditions: User owns private channel
Test Steps:
1. Open private channel
2. Click channel settings
3. Click "Invite Members"
4. Select user from workspace members
5. Click "Send Invitation"

Expected Result:
- Invitation sent to user
- Notification created
- User receives channel invite notification
- Channel remains private

TC-034: Accept Channel Invitation
Priority: High
Preconditions: User has pending channel invitation
Test Steps:
1. Open notifications panel
2. View channel invitation
3. Click "Accept"

Expected Result:
- User added to channel
- Channel appears in sidebar
- Can view channel messages
- Can send messages

TC-035: Archive Channel
Priority: Medium
Preconditions: User is channel owner
Test Steps:
1. Open channel
2. Click channel settings
3. Click "Archive Channel"
4. Confirm action

Expected Result:
- Channel archived (isArchived = true)
- Moved to "Archived" section
- Read-only access
- No new messages allowed
- Can be restored

TC-036: Delete Channel (Soft Delete)
Priority: Medium
Preconditions: User is channel owner
Test Steps:
1. Open channel settings
2. Click "Delete Channel"
3. Confirm deletion

Expected Result:
- Channel soft deleted (isDeleted = true)
- Moved to "Trash" section
- Not visible in main channel list
- Can be restored within retention period
- Can be permanently deleted

TC-037: Restore Archived Channel
Priority: Medium
Preconditions: Channel is archived
Test Steps:
1. Navigate to "Archived" section
2. Select archived channel
3. Click "Restore Channel"
4. Confirm action

Expected Result:
- Channel restored (isArchived = false)
- Moved back to active channels
- Full functionality restored
- Can send messages again

TC-038: Permanently Delete Channel
Priority: High
Preconditions: Channel is in trash
Test Steps:
1. Navigate to "Trash" section
2. Select deleted channel
3. Click "Delete Permanently"
4. Confirm action

Expected Result:
- Channel permanently deleted from database
- All messages deleted
- Cannot be restored
- Confirmation message displayed

TC-039: Leave Channel as Member
Priority: Medium
Preconditions: User is channel member (not owner)
Test Steps:
1. Open channel
2. Click channel settings
3. Click "Leave Channel"
4. Confirm action

Expected Result:

- User removed from channel
- Channel removed from sidebar
- Cannot access channel anymore
- Can be re-invited

TC-040: Remove Member from Channel
Priority: Medium
Preconditions: User is channel owner/admin
Test Steps:
1. Open channel settings
2. View members list
3. Select a member
4. Click "Remove Member"
5. Confirm action

Expected Result:
- Member removed from channel
- Member loses access immediately
- Channel removed from member's sidebar
- Notification sent to removed member

TC-041: Promote Member to Channel Admin
Priority: Medium
Preconditions: User is channel owner
Test Steps:
1. Open channel settings
2. View members list
3. Select a member
4. Click "Make Admin"
5. Confirm action

Expected Result:
- Member promoted to admin role
- Admin badge displayed
- Admin permissions granted
- Can manage channel members

TC-042: Demote Channel Admin to Member
Priority: Medium
Preconditions: User is channel owner, admin exists
Test Steps:
1. Open channel settings
2. View members list
3. Select an admin
4. Click "Remove Admin"
5. Confirm action

Expected Result:
- Admin demoted to member role
- Admin badge removed
- Admin permissions revoked
- Still remains as member

TC-043: Transfer Channel Ownership
Priority: Low
Preconditions: User is channel owner

Test Steps:
1. Open channel settings
2. Click "Transfer Ownership"
3. Select new owner from members
4. Confirm transfer

Expected Result:
- Ownership transferred
- Previous owner becomes admin
- New owner has full control
- Change reflected immediately

TC-044: View Channel Members List
Priority: Medium
Preconditions: User is channel member
Test Steps:
1. Open channel
2. Click channel name or info icon
3. View members list

Expected Result:
- All channel members displayed
- Shows display names and avatars
- Shows online status
- Shows member roles

TC-045: Create Channel with Duplicate Name
Priority: Low
Preconditions: Channel with name exists in workspace
Test Steps:
1. Try to create channel
2. Enter existing channel name
3. Complete form
4. Click "Create"

Expected Result:
- Channel created successfully
- Duplicate names allowed in same workspace
- Each channel has unique ID
- No conflict

Messaging Features

TC-046: Send Text Message in Channel
Priority: High
Preconditions: User is channel member
Test Steps:
1. Open channel
2. Type message: "Hello, team!"
3. Press Enter or click Send

Expected Result:
- Message sent successfully
- Message appears in chat window
- Timestamp displayed
- Sender name and avatar shown
- Other members see message in real-time

TC-047: Send Message with @Mention
Priority: High
Preconditions: User is channel member, other members exist
Test Steps:
1. Open channel
2. Type: "Hey @JohnDoe, check this out!"
3. Send message

Expected Result:
- Message sent with mention
- @JohnDoe highlighted in blue
- Notification sent to mentioned user
- Mentioned user sees "mentioned you" badge
- Message has blue left border for mentioned user

TC-048: Reply to Message
Priority: High
Preconditions: Messages exist in channel
Test Steps:
1. Hover over a message
2. Click "Reply" icon
3. Type reply: "Thanks for the update!"
4. Send message

Expected Result:
- Reply sent successfully
- Original message quoted in reply
- Reply linked to original message
- Thread context visible

TC-049: Edit Own Message
Priority: High
Preconditions: User has sent a message
Test Steps:
1. Hover over own message
2. Click "Edit" icon
3. Modify message text
4. Click "Save" or press Enter

Expected Result:
- Message updated successfully
- "Edited" label displayed
- Edit timestamp shown
- Updated message visible to all
- Original message replaced

TC-050: Delete Own Message (For Self)
Priority: Medium
Preconditions: User has sent a message
Test Steps:
1. Hover over own message
2. Click "Delete" icon
3. Select "Delete for me"
4. Confirm deletion

Expected Result:

- Message hidden for user only
- Other users still see message
- Message marked as hidden for user
- Cannot be recovered

TC-051: Delete Message for Everyone
Priority: High
Preconditions: User is message sender or channel admin
Test Steps:
1. Hover over message
2. Click "Delete" icon
3. Select "Delete for everyone"
4. Confirm deletion

Expected Result:
- Message deleted for all users
- Message marked as deleted (isDeleted = true)
- Shows "Message deleted" placeholder
- Cannot be recovered

TC-052: Pin Message to Channel
Priority: Medium
Preconditions: User is channel member
Test Steps:
1. Hover over important message
2. Click "Pin" icon
3. Confirm pin action

Expected Result:
- Message pinned successfully
- Pin icon displayed on message
- Message appears in pinned messages section
- All members can see pinned message

TC-053: React to Message with Emoji
Priority: Medium
Preconditions: Messages exist in channel
Test Steps:
1. Hover over a message
2. Click "Add Reaction" icon
3. Select emoji: ðŸ‘
4. Click emoji

Expected Result:
- Reaction added to message
- Emoji displayed below message
- Reaction count shown
- User's reaction highlighted
- Other users see reaction in real-time

TC-054: Remove Own Reaction
Priority: Low
Preconditions: User has reacted to a message
Test Steps:
1. View message with own reaction
2. Click on own reaction emoji

Expected Result:
- Reaction removed
- Emoji count decremented
- If count = 0, emoji removed
- Change visible to all users

TC-055: View Message History with Pagination
Priority: High
Preconditions: Channel has 100+ messages
Test Steps:
1. Open channel
2. Scroll to top of messages
3. Observe "Load More" behavior

Expected Result:
- Initial 50 messages loaded
- Scroll up loads previous messages
- Smooth pagination
- No duplicate messages
- Maintains scroll position

TC-056: Send Message with Multiple Mentions
Priority: Medium
Preconditions: Multiple users in channel
Test Steps:
1. Type message: "@John @Jane @Mike please review"
2. Send message

Expected Result:
- All three users mentioned
- All mentions highlighted
- Notifications sent to all three users
- Each user sees mention badge

TC-057: Send Long Message (1000+ characters)
Priority: Low
Preconditions: User in channel
Test Steps:
1. Type very long message (>1000 chars)
2. Send message

Expected Result:
- Message sent successfully
- Full message stored
- Message displayed with proper formatting
- Scrollable if needed
- No truncation

TC-058: Send Message with Special Characters
Priority: Low
Preconditions: User in channel
Test Steps:
1. Type message with special chars: "Test @#$%^&*() <script>"
2. Send message

Expected Result:
- Message sent successfully
- Special characters escaped/sanitized
- No XSS vulnerability
- Message displayed correctly

TC-059: Typing Indicator
Priority: Medium
Preconditions: Multiple users in channel
Test Steps:
1. User A starts typing
2. User B observes channel

Expected Result:
- "User A is typing..." displayed to User B
- Indicator appears within 1 second
- Indicator disappears after 3 seconds of inactivity
- Multiple users typing shown

TC-060: Message Read Receipts
Priority: Low
Preconditions: User sends message in channel
Test Steps:
1. Send message
2. Other users view message
3. Check read status

Expected Result:
- Read receipts tracked
- Can see who read message
- Read count displayed
- Updates in real-time

Direct Messaging

TC-061: Send DM Request to User
Priority: High
Preconditions: Two users in same workspace
Test Steps:
1. Click on user's name
2. Click "Send Message"
3. System creates DM request

Expected Result:
- DM request created with PENDING status
- Notification sent to recipient
- Request appears in recipient's notifications
- Sender can see pending status

TC-062: Accept DM Request
Priority: High
Preconditions: User has pending DM request
Test Steps:
1. Open notifications
2. View DM request
3. Click "Accept"

Expected Result:
- Request status changed to ACCEPTED
- DM conversation opened
- Both users can send messages
- Conversation appears in DM list

TC-063: Reject DM Request
Priority: Medium
Preconditions: User has pending DM request
Test Steps:
1. Open notifications
2. View DM request
3. Click "Reject"

Expected Result:
- Request status changed to REJECTED
- No DM conversation created
- Requester notified of rejection
- Can request again later

TC-064: Send Direct Message
Priority: High
Preconditions: DM conversation established
Test Steps:
1. Open DM conversation
2. Type message: "Hi, how are you?"
3. Send message

Expected Result:
- Message sent successfully
- Message appears in conversation
- Recipient receives notification
- Message delivered in real-time
- Unread badge shown to recipient

TC-065: View DM History
Priority: High
Preconditions: DM conversation with message history
Test Steps:
1. Open DM conversation
2. Scroll through messages
3. Load older messages

Expected Result:
- All messages displayed chronologically
- Pagination works smoothly
- Messages from both users shown
- Timestamps displayed correctly

TC-066: Mark DM as Read
Priority: Medium
Preconditions: User has unread DMs
Test Steps:
1. Open DM conversation with unread messages
2. View messages

Expected Result:
- Messages marked as read automatically
- Unread badge removed
- Read timestamp recorded
- Sender can see read status

TC-067: Edit DM Message
Priority: Medium
Preconditions: User has sent DM
Test Steps:
1. Hover over own DM
2. Click "Edit"
3. Modify message
4. Save changes

Expected Result:
- DM updated successfully
- "Edited" label shown
- Updated message visible to both users
- Edit timestamp recorded

TC-068: Delete DM for Self
Priority: Medium
Preconditions: DM conversation exists
Test Steps:
1. Hover over DM
2. Click "Delete"
3. Select "Delete for me"
4. Confirm

Expected Result:
- Message hidden for user only
- Other user still sees message
- Message marked as hidden
- Cannot be recovered

TC-069: React to DM
Priority: Low
Preconditions: DM conversation exists
Test Steps:
1. Hover over DM
2. Click "Add Reaction"
3. Select emoji
4. Confirm

Expected Result:
- Reaction added to DM
- Emoji displayed
- Other user sees reaction
- Can remove reaction

TC-070: Reply to DM
Priority: Medium
Preconditions: DM conversation exists
Test Steps:

1. Hover over DM
2. Click "Reply"
3. Type reply message
4. Send

Expected Result:
- Reply sent with context
- Original message quoted
- Reply linked to original
- Thread maintained

Notifications

TC-071: Receive Mention Notification
Priority: High
Preconditions: User mentioned in message
Test Steps:
1. Another user mentions you in channel
2. Observe notification bell icon

Expected Result:
- Notification received in real-time
- Bell icon shows unread count
- Notification type: MENTION
- Content shows who mentioned you
- Click navigates to message

TC-072: Receive Workspace Invitation Notification
Priority: High
Preconditions: User invited to workspace
Test Steps:
1. Workspace owner sends invitation
2. Check notification panel

Expected Result:
- Notification received immediately
- Type: WORKSPACE_INVITE
- Shows workspace name and inviter
- Accept/Reject buttons available
- Unread count incremented

TC-073: Receive Channel Invitation Notification
Priority: High
Preconditions: User invited to private channel
Test Steps:
1. Channel owner sends invitation
2. Check notification panel

Expected Result:
- Notification received in real-time
- Type: CHANNEL_INVITE
- Shows channel name and inviter
- Accept/Reject buttons available
- Click navigates to channel after accept

TC-074: Receive DM Notification
Priority: High
Preconditions: User receives new DM
Test Steps:
1. Another user sends DM
2. Observe notifications

Expected Result:
- Notification received immediately
- Type: DIRECT_MESSAGE
- Shows sender and message preview
- Unread badge on DM list
- Click opens DM conversation

TC-075: Mark Notification as Read
Priority: Medium
Preconditions: User has unread notifications
Test Steps:
1. Open notification panel
2. Click on a notification
3. Click "Mark as Read"

Expected Result:
- Notification marked as read
- Unread count decremented
- Notification styling changes
- Still visible in notification list

TC-076: View All Notifications
Priority: Medium
Preconditions: User has multiple notifications
Test Steps:
1. Click notification bell icon
2. View notification panel

Expected Result:
- All notifications displayed
- Sorted by date (newest first)
- Shows notification type icons
- Unread notifications highlighted
- Scrollable list

TC-077: Notification Badge Count
Priority: Medium
Preconditions: User has unread notifications
Test Steps:
1. Receive multiple notifications
2. Observe bell icon badge

Expected Result:
- Badge shows correct count
- Updates in real-time
- Maximum display (e.g., 99+)
- Decrements when marked read
- Disappears when count = 0

TC-078: Accept Invite from Notification
Priority: High
Preconditions: User has workspace/channel invite
Test Steps:
1. Open notification panel
2. View invitation notification
3. Click "Accept" button

Expected Result:
- Invitation accepted
- User added to workspace/channel
- Notification marked as read
- Success message displayed
- Workspace/channel appears in list

TC-079: Dismiss Notification
Priority: Low
Preconditions: User has notifications
Test Steps:
1. Open notification panel
2. Hover over notification
3. Click "Dismiss" or X icon

Expected Result:
- Notification removed from list
- Unread count decremented
- No other action taken
- Cannot be recovered

TC-080: Notification Persistence
Priority: Medium
Preconditions: User has notifications
Test Steps:
1. View notifications
2. Logout
3. Login again
4. Check notifications

Expected Result:
- Unread notifications persisted
- Still visible after re-login
- Count maintained
- No notifications lost

File Upload & Attachments

TC-081: Upload Image File
Priority: High
Preconditions: User in channel
Test Steps:
1. Click attachment icon
2. Select image file (PNG, JPG)
3. Confirm upload
4. Send message

Expected Result:
- Image uploaded to Cloudinary
- Image preview shown in message

- File URL stored in database
- Other users see image inline
- Click to view full size

TC-082: Upload Document File
Priority: High
Preconditions: User in channel
Test Steps:
1. Click attachment icon
2. Select document (PDF, DOCX)
3. Confirm upload
4. Send message

Expected Result:
- Document uploaded successfully
- File name and icon displayed
- Download link available
- File size shown
- Other users can download

TC-083: Upload File Size Limit
Priority: High
Preconditions: User in channel
Test Steps:
1. Try to upload file >10MB
2. Observe validation

Expected Result:
- Upload blocked
- Error message: "File too large (max 10MB)"
- File not uploaded
- User can select different file

TC-084: Upload Multiple Files
Priority: Medium
Preconditions: User in channel
Test Steps:
1. Click attachment icon
2. Select multiple files
3. Confirm upload
4. Send message

Expected Result:
- All files uploaded
- Multiple attachments in one message
- Each file shown separately
- All files accessible
- Proper file metadata

TC-085: View Attachment History
Priority: Low
Preconditions: Channel has file attachments
Test Steps:
1. Open channel info
2. Click "Files" tab
3. View all attachments

Expected Result:
- All channel attachments listed
- Sorted by date
- Shows file name, type, size
- Shows uploader name
- Click to download/view

User Profile & Presence

TC-086: Update Display Name
Priority: Medium
Preconditions: User logged in
Test Steps:
1. Click user profile icon
2. Click "Edit Profile"
3. Change display name to "John Smith"
4. Click "Save"

Expected Result:
- Display name updated
- New name shown everywhere
- Change reflected in messages
- Other users see new name
- Success message displayed

TC-087: Upload Profile Avatar
Priority: Medium
Preconditions: User logged in
Test Steps:
1. Open profile settings
2. Click "Upload Avatar"
3. Select image file
4. Crop/adjust if needed
5. Save changes

Expected Result:
- Avatar uploaded to Cloudinary
- New avatar displayed
- Avatar shown in messages
- Avatar shown in member lists
- Other users see new avatar

TC-088: View User Profile
Priority: Low
Preconditions: User in workspace
Test Steps:
1. Click on another user's name
2. View profile modal/page

Expected Result:
- Profile information displayed
- Shows display name, email
- Shows avatar
- Shows online status
- Shows joined date
- Option to send DM

TC-089: Online Status Indicator
Priority: High
Preconditions: Multiple users in workspace
Test Steps:
1. User A logs in
2. User B observes user list
3. User A logs out
4. User B observes again

Expected Result:
- Green dot when online
- Gray dot when offline
- "Last seen" timestamp when offline
- Updates in real-time
- Accurate status

TC-090: User Search Functionality
Priority: Medium
Preconditions: Multiple users in system
Test Steps:
1. Click search bar
2. Type user name: "John"
3. View search results

Expected Result:
- Matching users displayed
- Shows display name and avatar
- Shows online status
- Click to view profile or send DM
- Search is case-insensitive

Advanced Features

TC-091: WebSocket Connection Stability
Priority: High
Preconditions: User logged in
Test Steps:
1. Login to application
2. Disconnect internet briefly
3. Reconnect internet
4. Observe WebSocket reconnection

Expected Result:
- WebSocket reconnects automatically
- No data loss
- Messages sync after reconnection
- User notified of connection status
- Seamless experience

TC-092: Real-time Message Delivery
Priority: High
Preconditions: Two users in same channel
Test Steps:
1. User A sends message
2. User B observes channel (no refresh)

Expected Result:
- Message appears instantly for User B
- No page refresh needed
- Latency <1 second
- Message order maintained
- Consistent across all users

TC-093: Concurrent Message Sending
Priority: Medium
Preconditions: Multiple users in channel
Test Steps:
1. 5 users send messages simultaneously
2. Observe message order

Expected Result:
- All messages delivered
- Correct chronological order
- No messages lost
- No duplicate messages
- Proper conflict resolution

TC-094: Channel Unread Badge
Priority: Medium
Preconditions: User is member of multiple channels
Test Steps:
1. New message sent in Channel A
2. User viewing Channel B
3. Observe Channel A in sidebar

Expected Result:
- Unread badge appears on Channel A
- Badge shows message count
- Badge persists until channel opened
- Badge clears when channel viewed
- Accurate count

TC-095: DM Unread Badge
Priority: Medium
Preconditions: User has DM conversations
Test Steps:
1. Receive new DM
2. User viewing different conversation
3. Observe DM list

Expected Result:
- Unread badge on DM conversation
- Shows unread count
- Badge clears when DM opened
- Accurate count
- Updates in real-time

TC-096: Message Search in Channel
Priority: Low
Preconditions: Channel has many messages
Test Steps:
1. Open channel

2. Click search icon
3. Enter search term: "meeting"
4. View results

Expected Result:
- Matching messages displayed
- Highlights search term
- Shows message context
- Click to jump to message
- Search is case-insensitive

TC-097: Emoji Picker Functionality
Priority: Low
Preconditions: User composing message
Test Steps:
1. Click emoji icon in message input
2. Browse emoji categories
3. Select emoji
4. Send message

Expected Result:
- Emoji picker opens
- Categories displayed
- Search functionality works
- Selected emoji inserted
- Emoji displays correctly

TC-098: Message Timestamp Display
Priority: Low
Preconditions: Messages in channel
Test Steps:
1. View messages from different times
2. Observe timestamp format

Expected Result:
- Recent messages: "2 minutes ago"
- Today's messages: "10:30 AM"
- Yesterday: "Yesterday at 10:30 AM"
- Older: "Jan 15 at 10:30 AM"
- Hover shows full timestamp

TC-099: Browser Notification Permission
Priority: Low
Preconditions: User first login
Test Steps:
1. Login to application
2. Observe browser notification prompt
3. Grant permission

Expected Result:
- Browser asks for notification permission
- If granted, desktop notifications work
- If denied, in-app notifications still work
- Can change permission later
- Preference saved

TC-100: Application Performance with Large Data
Priority: High
Preconditions: User in workspace with 1000+ messages
Test Steps:
1. Open channel with large message history
2. Scroll through messages
3. Send new message
4. Observe performance

Expected Result:
- Smooth scrolling
- No lag or freezing
- Messages load quickly
- Pagination efficient
- Memory usage reasonable
- No browser crashes

Test Execution Summary

Test Environment
- **Frontend URL**: http://localhost:5173
- **Backend URL**: http://localhost:8080
- **Database**: PostgreSQL
- **Browsers**: Chrome, Firefox, Safari, Edge
- **Devices**: Desktop, Tablet, Mobile

Test Data Requirements
- Minimum 10 test user accounts
- 5 test workspaces
- 20 test channels (public and private)
- 100+ test messages
- Various file types for upload testing

Priority Distribution
- **High Priority**: 45 test cases (Critical functionality)
- **Medium Priority**: 40 test cases (Important features)
- **Low Priority**: 15 test cases (Nice-to-have features)

Expected Pass Rate
- **Target**: 95% pass rate
- **Acceptable**: 90% pass rate
- **Critical**: 100% pass rate for High Priority tests

Test Execution Checklist

- [ ] All test users created
- [ ] Test workspaces set up
- [ ] Test channels created
- [ ] Test data populated
- [ ] All browsers tested
- [ ] Mobile responsiveness checked
- [ ] Performance benchmarks met
- [ ] Security tests passed
- [ ] Accessibility verified
- [ ] Documentation updated

Defect Severity Levels

Critical: Application crash, data loss, security vulnerability
High: Major feature not working, blocking user workflow
Medium: Feature partially working, workaround available
Low: Minor UI issue, cosmetic problem

Sign-off

Prepared By: QA Team
Reviewed By: _____
Approved By: _____
Date: _____

End of Test Cases Document

This document contains 100 comprehensive functional test cases covering all major features and workflows of the ByteChat application. Each test case is designed to verify specific functionality and ensure the application meets quality standards.